The quartic-difference channel: completing the third-order inventory, with an honest endpoint-marginal verdict

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

U4 flagged the quartic-vertex second-order channel as the remaining third-order writeup. This note supplies it, completing the channel inventory for the SC-SCOPE lift. Scope: estimate grade (sup-kernel bounds on certified constants); the per-transfer refinement is registered, not performed.

2. The channel

Expanding the interaction about the condensate, the quartic vertex of the fluctuation is $V_4 = \int g_4 {:}\delta\varphi^4{:}$ with

$$g_4(x) \;=\; \frac{u_{\text{dressed}}}{4} \;+\; \frac{5v}{2}\,F(x)^2,$$

the F^2 piece coming from $(v/6)\binom{6}{4}F^2\delta\varphi^4$. Both $\mathcal{R}_H$ and competitors carry the constant part; the DIFFERENCE channel is driven by the pattern-dependent part, whose nonzero-transfer Fourier fibers are

$$\widehat{(F^2)}(t)\Big|_{t\neq 0} \;=\; \frac{w_t}{\lambda'}$$

(the chain's Parseval identity, main note v1.8) — the channel rides EXACTLY the certified transfer set. Its contribution,

$$\delta F_4^{(t\neq 0)} \;=\; -12\left(\frac{5v}{2}\right)^2\frac{1}{\lambda'^2}\sum_{t\neq 0} w_t^2\,\widehat{G^4}(t), \qquad \widehat{G^4} = G{*}G{*}G{*}G,$$

$(-\frac{1}{2}\cdot 4! = -12$; normal-ordered pairs), is competitor-helping and absent for $\mathcal{R}_H$ at $t \neq 0$. The $t = 0$ (uniform dressing-shift) part has its self-contractions resummed by the gap/condensate equations exactly as in U6 (normal-ordering); its residual is the convention remainder already bounded in U11.

3. Size relative to the certified second-order budget

Per transfer, the ratio to the second-order weight $J(t)/(4(1-a_0))$ is

$$\mathcal{R}(t) \;=\; 12\,\frac{(5v/2)^2}{\lambda'^2}\,\widehat{G^4}(t)\,\frac{4(1-a_0)}{J(t)}$$

$$\leq\; 12\cdot\frac{65.61}{64.88}\cdot\big(J(0)M_R^2\big)\cdot\frac{3.92}{J(2q_0)} \;=\; 12.14\times 3.47\times 10^{-3}\times 37.7 \;\approx\; 1.59,$$

using the Young bound $\widehat{G^4} \le \|G{*}G\|_\infty \|G{*}G\|_1 = J(0)M^2$ and the worst realized $J(t) \ge J(2q_0) = 0.104$. The channel therefore INFLATES the certified second-order budget by at most $1{+}1.59 \approx \times 2.6$ under sup-kernels, I-independently. Against the hardened margins:

	$I = 4{\times}10^{-4}$	$I = 10^{-3}$	$I = 2{\times}10^{-3}$
STEP-5B margin floor	$\times$59.4	$\times$8.8	$\times$2.6
after $\times$2.6 inflation	$\times$22.8	$\times$3.4	$\times$1.0 (MARGINAL)

4. Honest verdict and the completed inventory

- This channel EXCEEDS the cubic sunset as the dominant third- order threat under sup-kernels (the sunset's endpoint ratio was $\sim \times 1.34$ dressed; this channel alone consumes the entire $\times 2.6$ at the endpoint in the worst-kernel reading).
- The verdict is a statement about SUP-KERNEL crudeness, not the selection: $\widehat{G^4}(t)$ at the realized transfers ($|t| \ge q_0$-scale) sits below its Young bound, and $J(t)$ at the short transfers (where $\widehat{G^4}$ is largest) is $J(0)$-scale not $J(2q_0)$ — the ratio's worst case pairs the two extremes incompatibly. The per-transfer evaluation of $\widehat{G^4}(|t|)$ on the realized chord set (the same five chords the suite tabulates for J) is registered as ONE extension of the U14 draft script and is expected to reduce $\mathcal{R}$ substantially at every realized t; its outcome decides the endpoint.
- THIRD-ORDER INVENTORY (complete at estimate grade): cubic sunset — positive at all anchors pending M-ENDPOINT (U4/U7); tadpole — absent identically (U6); quartic-difference — anchor-comfortable, ENDPOINT-MARGINAL pending per-transfer kernels (this note). The SC-SCOPE lift at the endpoint requires the per-transfer programme for BOTH surviving channels; at the production anchor the lift is supported by every channel's estimate.

5. Numerical verification

NO machine execution this session. New estimate arithmetic on certified constants: $(5v/2)^2 = 65.61$; $\lambda'^2 = 64.88$; $J(0)M_R^2 = 3.47 \times 10^{-3}$; $4(1-a_0) = 3.92$ (anchor) with the I-independence of $\mathcal{R}$ noted (a_0 varies the ratio by $< 8\%$ across the band); the inflation table. Registered: the $\widehat{G^4}(|t|)$ chord-set evaluation appended to the U14 script (S3 extension) with asserts per chord.

Quantitative sanity check (mandatory): dimensional — $\widehat{G^4}$ carries one more G-convolution than $\widehat{G^3}$, hence one more M-factor in the Young chain ($J(0)M^2$ vs $J(0)M$), consistent term-by-term; magnitude — the channel's coupling ratio $12(5v/2)^2/\lambda'^2 = 12.14$ exceeds the sunset's u_{eff}^2-based weight because the sextic coupling $v = 3.24$ is large at this operating point — the finding that the QUARTIC channel dominates is therefore not surprising in retrospect and is the kind of fact the inventory exists to surface; limit case — at $v \to 0$ the channel's pattern part vanishes (g_4 becomes constant) and the inventory reduces to the sunset alone, consistently; sign-direction — the channel is competitor-helping and is applied against the margin throughout.

6. Devil's-advocate (three objections; CLAUDE.md 6.3)

(α) *"Lemma H's sextic fold ($\varepsilon_4 \le 0.16$) may already count part of this channel — the inflation could double-count."* VALID-with-mitigation: Lemma H folds the sextic VERTEX corrections to the transfer weights ($5vA^4$-class transfers, the G2 bookkeeping); this channel is the quartic-vertex-squared propagator loop ($\widehat{G^4}$ kernel) — a different contraction topology. The disjointness is asserted here at estimate grade and its one-paragraph verification (contraction-topology audit) is folded into the registered per-transfer extension.

(β) *"The $\times 2.6$ inflation multiplies the BUDGET, but the realized second-order consumption of actual patterns is far below budget — the inflation framing overstates the threat."* DISMISSED as framing: the inflation is applied to the PROVEN bound because that is what a lift must preserve; the note's endpoint verdict explicitly says 'marginal under sup-kernels, per-transfer decides' rather than 'fails'. The realized-consumption observation is true and is why the per-transfer programme is expected to succeed — recorded in section 4.

(γ) *"The reabsorption of the $t = 0$ part leans on U6's normal-ordering, which is itself T3."* CORRECT and tracked: the dependency is explicit (U6's R-U6-1 alignment writeup covers both vertices — the normal-ordering statement is vertex-degree- agnostic), and this note inherits T2/T3 accordingly. Nothing here is graded above its inputs.

7. Result footer

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Result ID:                B5-BEYOND-LAYER-BOUND / quartic-difference-
                          channel (unit U15)
Precise statement:        ESTIMATE (T2): the quartic-difference
                          channel's t != 0 part rides the certified
                          transfer set (fibers w_t / lam') with
                          per-transfer ratio R(t) <= ~1.59 under
                          sup-kernels => budget inflation <= x2.6,
                          I-independent: anchor x22.8 / middle x3.4 /
                          endpoint x1.0 MARGINAL. t = 0 part reabsorbed
                          (U6 normal-ordering, vertex-degree-agnostic).
                          THIRD-ORDER INVENTORY COMPLETE at estimate
                          grade: sunset positive (pending M-ENDPOINT),
                          tadpole absent, quartic-difference endpoint-
                          marginal (pending per-transfer G^4).
Scope:                    SC-SCOPE lift programme; estimate grade;
                          amended class, three anchors.
Dependencies:             U4/U6/U7 (channel framework), Parseval
                          identity (main v1.8), Lemma H disjointness
                          (asserted, audit registered), certified
                          constants.
Evidence grade:           ESTIMATE (T2); machine asserts NOT executed
                          (registered as U14-script S3 extension).
Reproduction command:     (registered) G^4(|t|) chord-set evaluation
                          appended to codes/vacuum/
                          t6_mainline_useries_checks.py.
Expected output:          per-chord G^4 values with R(t) < 1 expected
                          at all realized chords; contraction-topology
                          audit paragraph; JSON artefact.
Falsification gate:       Per-transfer R(t) >= 1 at a realized chord
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	(would mean the lift needs the joint second+third-order inequality re-derivation at the endpoint -- honest negative pre-registered); Lemma-H overlap found in the topology audit (would require subtracting the double-count, IMPROVING the verdict).
Tier before / after:	No tier action; SC-SCOPE remains a named U1 hypothesis, now with its full third-order landscape mapped.
No-overclaim statement:	Nothing here lifts SC-SCOPE or threatens the U1 theorem (second cumulant by hypothesis). The endpoint x1.0 is a sup-kernel artifact statement, pre-registered as such. PDF build + linter PENDING operator-side.
Next required action:	Per-transfer G^4 chord evaluation (U14-script extension) + contraction-topology audit; then the assembled third-order inequality IF the operator directs the SC-SCOPE lift. Mainline continues (U16: consolidation).